

# Spatially regularized low-rank tensor approximation for accurate and fast tractography

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## ABSTRACT

Tractography based on diffusion Magnetic Resonance Imaging (dMRI) is the prevalent approach to the in vivo delineation of white matter tracts in the human brain. Many tractography methods rely on models of multiple fiber compartments, but the local dMRI information is not always sufficient to reliably estimate the directions of secondary fibers. Therefore, we introduce two novel approaches that use spatial regularization to make multi-fiber tractography more stable. Both represent the fiber Orientation Distribution Function (fODF) as a symmetric fourth-order tensor, and recover multiple fiber orientations via low-rank approximation. Our first approach computes a joint approximation over suitably weighted local neighborhoods with an efficient alternating optimization. The second approach integrates the low-rank approximation into a current state-of-the-art tractography algorithm based on the unscented Kalman filter (UKF). These methods were applied in three different scenarios. First, we demonstrate that they improve tractography even in high-quality data from the Human Connectome Project, and that they maintain useful results with a small fraction of the measurements. Second, on the 2015 ISMRM tractography challenge, they increase overlap, while reducing overreach, compared to low-rank approximation without joint optimization or the traditional UKF, respectively. Finally, our methods permit a more comprehensive reconstruction of tracts surrounding a tumor in a clinical dataset. Overall, both approaches improve reconstruction quality. At the same time, our modified UKF significantly reduces the computational effort compared to its traditional counterpart, and to our joint approximation. However, when used with ROI-based seeding, joint approximation more fully recovers fiber spread.

## 1. Introduction

Diffusion Magnetic Resonance Imaging (dMRI) (Bihan et al., 1986) is a vital method for non-invasive imaging of the human brain, with dMRI-based tractography providing the unique ability to reconstruct the trajectories of the major white matter tracts in vivo (Jeurissen et al., 2019). It is widely used in surgery planning (Wilke et al., 2018; Yang et al., 2021), in large scientific studies (Sotiropoulos et al., 2013; Tobisch et al., 2018), where it gives information about brain structures and neurological diseases, and in the field of “connectomics”, which uses graph theory to study complex brain networks (Bullmore and Sporns, 2009).

Among the many algorithmic approaches to dMRI tractography, which more recently included global (Reisert et al., 2011) and machine learning based techniques (Poulin et al., 2019), streamline-based meth-

ods were the first (Basser et al., 2000; Mori et al., 1999), and remain the most widely used ones. They build on mathematical models that recover local fiber directions from dMRI measurements, which leads to an ill-posed inverse problem (Tournier et al., 2007). Over the past decade, several methods have been introduced to recover these local directions. The earliest approaches have used diffusion tensor imaging (Basser et al., 1994) and could only recover one direction per voxel. However, this turned out to be insufficient for the complex white matter structures that are found in large parts of the human brain (Behrens et al., 2007; Jeurissen et al., 2012). Newer approaches therefore recover multiple directions per voxel using mathematical models like the ball-and-stick method (Behrens et al., 2007), constrained spherical deconvolution (Tournier et al., 2007), or the low-rank approximation of higher-order tensors (Ankele et al., 2017; Schultz and Seidel, 2008).

**Abbreviations:** CSD, Constrained spherical deconvolution; dMRI, Diffusion Magnetic Resonance Imaging; DTI, Diffusion Tensor Imaging; fODF, fiber Orientation Distribution Function; UKF, Unscented Kalman filter; HCP, Human Connectome Project; OL, Overlap; OR, Overreach; ROI, Region Of Interest.

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Unfortunately, the higher complexity of these models also increases their susceptibility to measurement noise and artifacts. This poses particular challenges when applying them to data that has been acquired in a clinical setting, where scan times have to be kept short, and the available scanner hardware and corresponding licenses may pose additional limitations. Errors that are introduced at the voxel level accumulate and amplify during tractography, since generating a single streamline involves tracking through a large number of voxels. This leads to false as well as missing connections.

Spatial regularization is one way to increase the stability of tractography, by estimating fiber orientations not just based on the local measurements, but also accounting for information from neighboring voxels belonging to the same tract. One popular algorithm that achieves this is based on the unscented Kalman filter (UKF) (Cheng et al., 2015; Malcolm et al., 2010), which maintains an internal state vector including the fiber parameters while tracking, and updates them in each step, rather than estimating them independently from the measurements at each location. This approach has been shown to reduce the angular error in crossing fiber tractography, and has been used for neurosurgical planning (Chen et al., 2016) as well as in scientific studies (Dalamagkas et al., 2020; Wu et al., 2020).

Representing fiber Orientation Distribution Functions (fODFs) as symmetric higher-order tensors and extracting the principal directions for fiber tractography via low-rank approximation (Schultz and Seidel, 2008) has previously been shown to yield fast and accurate per-voxel estimates, due to its relatively good numerical conditioning (Ankele et al., 2017), but has not been combined with spatial regularization. In this work, we compare two specific approaches for doing so: The first one accounts for spatial regularity by jointly approximating all tensors in a local neighborhood, with weights that ensure that only voxels from the same tracts are considered. The second one modifies the UKF to work with higher-order tensor fODFs instead of the previously used multi-tensor model (Cheng et al., 2015; Malcolm et al., 2010). This increases computational efficiency by replacing the often very high-dimensional diffusion-weighted signal space with the much lower-dimensional fODFs. Simultaneously, it permits more flexible modeling of the single fiber response function. When combined with fODFs from multi-shell multi-tissue deconvolution (Ankele et al., 2017; Jeurissen et al., 2014), it also eliminates contributions from gray matter and cerebrospinal fluid.

## 2. Materials and methods

### 2.1. Low-Rank tensor approximation

Constrained spherical deconvolution (CSD) determines a fiber orientation distribution function (fODF), i.e., a mapping from the sphere to  $\mathbb{R}_+$  which captures the fraction of fibers in any direction (Tournier et al., 2007). The local maxima of the fODF are then used as estimates of principal fiber orientations. Traditionally, fODFs are expressed in a spherical harmonics basis, and a sufficiently high order (e.g., eight) is required to resolve multiple fibers as separate maxima.

Our work builds on a variation of constrained spherical deconvolution (Ankele et al., 2017), which represents the fODF as a symmetric fourth-order tensor  $\mathcal{T}$  and estimates  $r$  fiber directions from it via rank- $r$  approximation

$$\mathcal{T}^{(r)} = \sum_{i=1}^r \alpha_i \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i, \quad (1)$$

where the scalar  $\alpha_i \in \mathbb{R}_+$  denotes the volume fraction of the  $i$ th fiber,  $\mathbf{v}_i \in \mathbb{S}^2$  its direction, and  $\otimes$  the outer product. The parameters are chosen to minimize the Frobenius norm of the residual  $\|\mathcal{T}^{(r)} - \mathcal{T}\|$ .

The low-rank approximation permits separating crossing fibers even if they are not represented by distinct local fODF maxima. Prior work (Ankele et al., 2017) has shown that the angular resolution of fourth-order tensor approximation already exceeds the traditional peak finding

with order-eight spherical harmonics. At the same time, it reduces the number of fODF coefficients from 45 to 15, and leads to much better numerical conditioning. So far, low-rank approximation has performed for each voxel independently. Our current work further improves it by applying spatial regularization to it.

### 2.2. Joint low-rank approximation

Our first approach achieves spatial regularization by considering a set of  $n$  fourth-order fODF tensors  $\mathcal{T}_j$ ,  $j = 1, \dots, n$ , from the voxel of interest and its spatial neighborhood, along with normalized weighting factors  $w_j \in \mathbb{R}_+$ ,  $\sum_j w_j = 1$  that reflect the relative importance of voxel  $j$ , and defining their joint low-rank approximation as the rank- $r$  tensor that minimizes the objective function

$$f(\lambda_1, \dots, \lambda_r, \mathbf{v}_1, \dots, \mathbf{v}_r) = \sum_j w_j \left\| \mathcal{T}_j - \sum_{i=1}^r \lambda_i^2 \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \right\|^2 \quad (2)$$

with unit vectors  $\mathbf{v}_i \in \mathbb{S}^2$ ,  $i = 1, \dots, r$ , and scalar  $\lambda_i \in \mathbb{R}$  that reparameterizes  $\alpha_i = \lambda_i^2$  from Eq. (1) to account for the non-negativity constraint.

We propose to optimize Eq. (2) iteratively, alternating over  $i$ . The main difference that we found empirically between the solution from such an alternating scheme and a joint optimization of all variables is that the latter sometimes led to pairs of near-parallel vectors, which might be attributed to fiber bending or fanning, but are less relevant for estimating crossings. At the same time, alternating over  $i$  simplifies the problem and allows us to perform the optimization very efficiently. Therefore, we update the  $i$ th rank-one term which is based on the residual

$$\mathcal{R}_j^{(i)} := \mathcal{T}_j - \sum_{k \neq i} \lambda_k^2 \mathbf{v}_k \otimes \mathbf{v}_k \otimes \mathbf{v}_k \otimes \mathbf{v}_k \quad (3)$$

that results from subtracting all remaining rank-one terms from  $\mathcal{T}_j$ . Then, the original problem can be rewritten as

$$\begin{aligned} \tilde{f}(\lambda_i, \mathbf{v}_i) &= \sum_j w_j \left\| \mathcal{R}_j^{(i)} - \lambda_i^2 \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \right\|^2 \\ &= \sum_j w_j \left( \left\| \mathcal{R}_j^{(i)} \right\|^2 - 2 \left\langle \mathcal{R}_j^{(i)}, \lambda_i^2 \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \right\rangle \right. \\ &\quad \left. + \left\| \lambda_i^2 \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \right\|^2 \right) \\ &= \sum_j w_j \left( \left\| \mathcal{R}_j^{(i)} \right\|^2 - 2 \lambda_i^2 \mathcal{R}_j^{(i)}(\mathbf{v}_i) + \lambda_i^4 \right), \end{aligned} \quad (4)$$

where we have used that  $\|\mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i \otimes \mathbf{v}_i\|^2 = 1$  and  $\mathcal{R}_j^{(i)}(\mathbf{v}_i)$  is the homogeneous form of  $\mathcal{R}_j^{(i)}$ . Since the first summand is constant, the above is equal to maximizing

$$g(\lambda_i, \mathbf{v}_i) := 2 \lambda_i^2 \mathcal{B}^{(i)}(\mathbf{v}_i) - \lambda_i^4 \quad \text{subject to} \quad \|\mathbf{v}_i\| = 1 \quad (5)$$

with  $\mathcal{B}^{(i)} := \sum_j w_j \mathcal{R}_j^{(i)}$  and  $\sum_j \mathcal{R}_j^{(i)}(\mathbf{v}) = \left( \sum_j \mathcal{R}_j^{(i)} \right)(\mathbf{v})$ . Taking the derivative of Eq. (5) with respect to  $\lambda_i$ , we can derive that

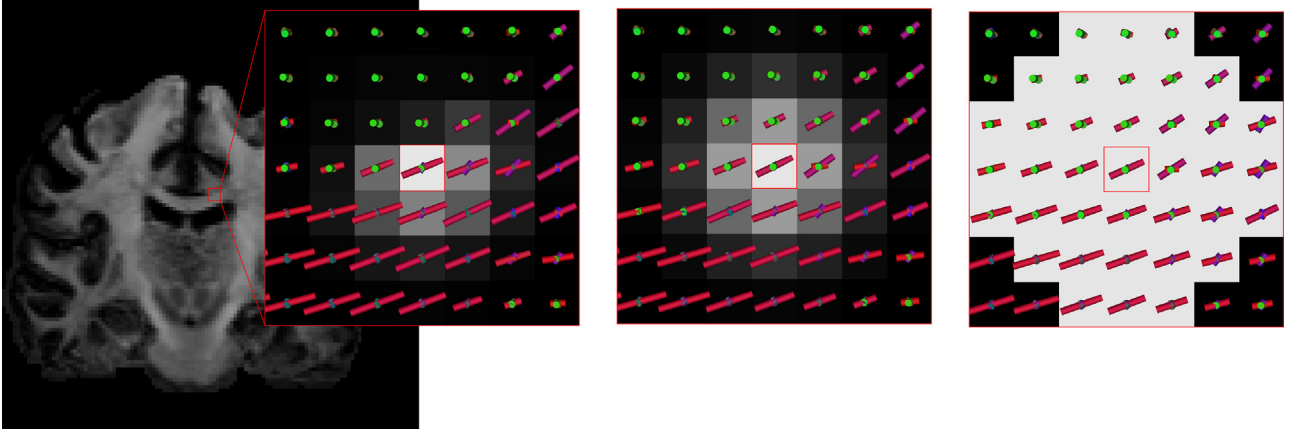
$$\lambda_i = \pm \sqrt{\mathcal{B}^{(i)}(\mathbf{v}_i)}. \quad (6)$$

This allows us to rewrite Eq. (5) as the constrained problem

$$\arg \max g(\lambda_i, \mathbf{v}_i) = \left( \mathcal{B}^{(i)}(\mathbf{v}_i) \right)^2 \quad \text{subject to} \quad \|\mathbf{v}_i\| = 1, \quad (7)$$

which we optimize using gradient ascent on the unit sphere, with Armijo step size control (Armijo, 1966).

Summarizing the above, we have shown that optimizing the joint low-rank approximation with an alternating gradient descent approach amounts to simply operating on the weighted sum of all involved tensors  $\mathcal{T}_j$ . This sum is easily computed in  $O(n)$  time, and the computational effort for the optimization itself becomes independent of the number of tensors that are involved. Therefore, the proposed approach remains efficient even if we account for 3D neighborhoods of substantial size. We will now introduce a suitable definition of the weights  $w_j$ .



**Fig. 1.** An illustration of three different weighting schemes, applied in a neighborhood of radius 4 mm around the center voxel, which is marked by a red square. Weights are color coded in grayscale, with brighter colors representing higher weights. Line glyphs indicate the results of joint low-rank approximation with the different weighting schemes. Left: Proposed approach that combines distance and fODF similarity weighting with  $\sigma_1 = 0.35$  and  $\sigma_2 = 2$ . Middle: Distance weighting with  $\sigma_2 = 2$ . Right: Uniform weighting. The proposed weights lead to smooth fiber directions within tracts, but preserve borders between them. Uniform and Gaussian weights are unable to separate the tracts. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

### 2.3. Weights and neighborhoods

Neighborhood size and voxel weights in our joint approximation have to be chosen carefully to achieve spatial regularization without introducing an undesirable blurring of tracts. This is illustrated in the rightmost part of Fig. 1, where applying joint approximation naïvely introduces a strong mixing between the corpus callosum (red line segments) and the adjacent cingulum bundle (green segments, mostly perpendicular to the image plane).

Therefore, we weight the contribution of each voxel based on its distance to the center, and the similarity of its fODF. The similarity of two fODFs  $\mathcal{T}_1, \mathcal{T}_2$  is defined as

$$p(\mathcal{T}_1, \mathcal{T}_2) := \exp\left(-\frac{\|\mathcal{T}_1 - \mathcal{T}_2\|_F^2}{\sigma_1^2}\right), \quad (8)$$

where  $\|\cdot\|_F$  denotes the Frobenius norm and  $\sigma_1$  a scaling factor. A second factor gives closer voxels a higher impact than voxels that are far away. It is defined as

$$d(\mathbf{x}_1, \mathbf{x}_2) := \exp\left(-\frac{\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2}{\sigma_2^2}\right), \quad (9)$$

where  $\mathbf{x}_i$  is the spatial position of  $\mathcal{T}_i$  and  $\sigma_2$  is another scaling factor. Similar to bilateral filtering (Tomasí and Manduchi, 1998), both measures are combined to

$$p_{\mathcal{T}, \mathbf{x}}(\mathcal{T}_i, \mathbf{x}_i) := \frac{p(\mathcal{T}, \mathcal{T}_i) \cdot d(\mathbf{x}, \mathbf{x}_i)}{\sum_j p(\mathcal{T}, \mathcal{T}_j) \cdot d(\mathbf{x}, \mathbf{x}_j)}, \quad (10)$$

for an fODF  $\mathcal{T}$  at spatial position  $\mathbf{x}$  and its neighbours  $\mathcal{T}_j, \mathbf{x}_j$  for  $j \in \{1, \dots, n\}$ . Then, we can compute the corresponding weighted sum of fODFs for the joint approximation.

For adequately chosen scaling factors  $\sigma_1, \sigma_2$ , the tract boundaries remain sharp, but regularization is effective within the tracts and reduces the tracking uncertainty. This is illustrated in the left detail view in Fig. 1. For comparison, the central example uses a purely distance-based weighting, which is shown to be less effective at keeping the tracts separated than the combination of both factors.

While tuning  $\sigma_1$  and  $\sigma_2$  to specific datasets, and potentially even to specific tracts, might optimize results further, having to do so would pose a significant burden on the user. Therefore, we report results that can be achieved by automatically setting them to reasonable defaults. The default value of  $\sigma_1$  is motivated by observing that setting it much larger than the fODF differences in local neighborhoods would lead to meaningless weights  $p(\mathcal{T}_1, \mathcal{T}_2) \approx 1$ , while much smaller values would

lead to  $p(\mathcal{T}_1, \mathcal{T}_2) \approx 0$ . We avoid these extremes by adapting  $\sigma_1$  based on neighborhoods with a radius of at least 3mm and at least 27 nearest voxels. In particular, we set  $\sigma_1$  to the average fODF distances within those neighborhoods

$$\sigma_1 := \frac{1}{m} \sum_{\mathbf{x}} \frac{1}{n} \sum_j \|\mathcal{F}_{\mathbf{x}} - \mathcal{F}_{B(\mathbf{x})_j}\|, \quad (11)$$

where  $\mathcal{F}$  denotes the fODFs which are indexed by the voxel  $\mathbf{x}$  they belong to,  $B(\mathbf{x})$  the neighborhood with size  $n$  of voxel  $\mathbf{x}$ , and  $m$  the number of fODFs within the white matter mask. Similarly, we account for the variability in resolution among dMRI datasets by setting  $\sigma_2$  to the average voxel edge length, to ensure that regularization remains effective independently of image resolution.

### 2.4. UKF with low-rank signal model

Our second approach integrates the low-rank tensor approximation as a signal model into an unscented Kalman filter (UKF), as it was first employed for tractography by Malcolm et al. (2010). Here, spatial regularization is achieved by maintaining a state vector of the fiber configuration in the previous tracking step, and updating it in each step in a way that accounts for the relative uncertainty in the previous estimate, as well as the local uncertainty.

Specifically, our state vector contains the directions  $\mathbf{v}_i \in \mathbb{S}^2$ , and square roots of volume fractions  $\lambda_i \in \mathbb{R}$  of up to two crossing fibers from Eq. (2),

$$\mathbf{x}_t = [\lambda_1, \mathbf{v}_{1,1}, \mathbf{v}_{1,2}, \mathbf{v}_{1,3}, \lambda_2, \mathbf{v}_{2,1}, \mathbf{v}_{2,2}, \mathbf{v}_{2,3}] \in \mathbb{R}^8, \quad (12)$$

where  $t \in \mathbb{N}$  denotes the step, and absence of a second fiber would be indicated by a corresponding  $\lambda_i \approx 0$ . Previous work compared UKF tractography with two- and three-fiber models, and found that “either model resulted in generally the same pathways” (Malcolm et al., 2010). Since experiments with our low-rank UKF led to the same conclusion, we only present results from the simpler two-fiber model.

Our implementation follows the same basic steps as a traditional UKF tractography (Malcolm et al., 2010). We initialize the UKF state with a rank-2 approximation of the nearest neighbor of the seed point as  $\mathbf{x}_0$ , and set the initial covariance  $\mathbf{P}_0 \in \mathbb{R}^{8 \times 8}$  to the identity, since we have no information so far. The UKF involves the computation of sigma points, for which we first define associated weights  $\omega_i \in \mathbb{R}$  as

$$\omega_0 = \frac{\kappa}{d + \kappa} \quad \text{and} \quad \omega_i = \frac{1}{2(d + \kappa)} \quad \text{for } i \in \{1, \dots, 2d\}, \quad (13)$$

where  $d = 8$  is the dimension of our state vector and  $\kappa \in \mathbb{R}_+$  is a scaling parameter which is set to  $\kappa = 0.03$  by default. For a given step  $t \in \mathbb{N}$ , we build  $2d + 1$  sigma points  $\mathbf{X}_t := \{\chi_i\}_{i \in \{0, \dots, 2d\}} \subset \mathbb{R}^d$  with associated convex weights  $\omega_i$  for  $i \in \{0, \dots, 2d\}$  around the current state  $\mathbf{x}_t$ . Therefore, we first calculate the Cholesky decomposition  $\mathbf{U}$  of  $(d + \kappa)\mathbf{P}_t$ , which we then use to calculate the sigma points as

$$\chi_0 = \mathbf{x}_t \quad \text{and} \quad \chi_{i+1} = \mathbf{x}_t + \mathbf{U}_{:,i} \quad \text{and} \quad \chi_{i+d+1} = \mathbf{x}_t - \mathbf{U}_{:,i} \quad \text{for} \\ i \in \{0, \dots, d-1\}. \quad (14)$$

Next, we apply the transition function  $f$  to the newly generated sigma points. Since we assume local smoothness, the transition function is set to the identity, i.e.,  $\mathbf{X}_{t+1|t} := f(\mathbf{X}_t) = \mathbf{X}_t$ . From the predicted sigma points, we calculate the mean state

$$\bar{\mathbf{x}}_{t+1|t} = \sum \omega_i \chi_i \quad (15)$$

and the covariance

$$\mathbf{P}_{xx} = \sum \omega_i (\chi_i - \bar{\mathbf{x}}_{t+1|t})(\chi_i - \bar{\mathbf{x}}_{t+1|t})^T + \mathbf{Q}, \quad (16)$$

where  $\mathbf{Q}$  is the injected process noise, which ensures a non-null spread of sigma points.

The key difference towards the UKF tractography algorithm by Malcolm et al. is our signal model. In particular, instead of predicting the raw dMRI signals in each step as a mixture of diffusion tensors, we predict fourth-order tensor fODFs  $\gamma_i \in \mathbb{R}^{15}$ ,  $i \in \{0, \dots, 2d\}$  via Eq. (1). Based on them, we compute the predicted fODF mean  $\bar{\gamma}_{t+1|t}$  and its covariance  $\mathbf{P}_{yy} \in \mathbb{R}^{15 \times 15}$  with injected measurement noise  $\mathbf{R}$  in the same manner as the predicted state mean and its covariance were calculated above. Further, the cross correlation between predicted state and measurement is calculated

$$\mathbf{P}_{xy} = \sum \omega_i (\chi_i - \bar{\mathbf{x}}_{t+1|t})(\gamma_i - \bar{\gamma}_{t+1|t})^T \quad (17)$$

and used to calculate the Kalman gain

$$\mathbf{K} = \mathbf{P}_{xy} \mathbf{P}_{yy}^{-1} \in \mathbb{R}^{8 \times 15}. \quad (18)$$

Finally, the Kalman gain is used to calculate an updated state vector

$$\mathbf{x}_{t+1} = \bar{\mathbf{x}}_{t+1|t} + \mathbf{K}(\mathbf{y}_t - \bar{\gamma}_{t+1|t}), \quad (19)$$

where  $\mathbf{y}_t \in \mathbb{R}^m$  is the trilinearly interpolated fODF at step  $t$  and the covariance matrix

$$\mathbf{P}_{t+1} = \mathbf{P}_{xx} - \mathbf{K} \mathbf{P}_{yy} \mathbf{K}^T. \quad (20)$$

The details of how this newly generated state are used for tractography will be discussed in the next subsection.

An immediate practical benefit of our modification is that it reduces the computational complexity of the original UKF. In particular, the signal vectors  $\gamma_i$  are now only 15-dimensional, while their dimension matches the number of diffusion-weighted measurements in the original formulation. In the HCP data considered below, this amounts to 270 dimensions when considering all three shells with 90 measurements each. This difference is particularly relevant when inverting the matrix  $\mathbf{P}_{yy}$ .

Tuning the parameters of the UKF approach on an example subject from the HCP dataset that is described below led us to set  $\mathbf{Q}$  to a diagonal matrix with entries  $\mathbf{Q}_{\lambda_i} = 0.2$  and  $\mathbf{Q}_{\Theta_i} = 0.03$  for  $i = 1, 2$ . Further, we set the diagonal entries of  $\mathbf{R}$  to  $\mathbf{R}_{i,i} = 0.002$  for  $i \in \{0 \dots 14\}$ .

## 2.5. Probabilistic streamline-based tractography

In either approach, we track iteratively in both directions, with a fixed stepsize of 0.7 mm, until a stopping criterion is reached. At each step, we have to decide in which direction we should continue.

If the joint low-rank approximation is used, we determine a set of possible directions  $\mathbf{v}_i$ ,  $i = 1, \dots, r$ , by optimizing Eq. (2). To speed up convergence and to help avoid suboptima, we initialize with the parameters from the previous tracking step. In agreement with previous work on low-rank approximation based tractography (Ankele et al., 2017), we set  $r = 3$  to capture up to three crossing fiber directions.

If the UKF is used, the updated state vector provides a set of possible directions. In this case, in agreement with Malcolm et al. (2010), we only model up to two fibers given that, by construction of the UKF, one of them is guaranteed to be the current tracking direction.

Among the candidates  $\hat{\mathbf{w}}_i$  with length  $\hat{\lambda}_i$ , we select the next tracking direction probabilistically, by assigning the probabilities

$$p_{\lambda, \mathbf{w}}(\hat{\lambda}_i, \hat{\mathbf{w}}_i) := \frac{\mathbf{1}_{\{\Theta_i < \frac{1}{3}\pi\}} \exp\left(-\frac{\|\lambda - \hat{\lambda}_i\|}{2}\right) \cos\left(\left(\frac{3}{\sqrt{2\pi}}\Theta_i\right)^2\right)^6}{\sum_j \mathbf{1}_{\{\Theta_j < \frac{1}{3}\pi\}} \exp\left(-\frac{\|\lambda - \hat{\lambda}_j\|}{2}\right) \cos\left(\left(\frac{3}{\sqrt{2\pi}}\Theta_j\right)^2\right)^6},$$

based on the previous tracking direction  $\mathbf{w}$  with length  $\lambda$ , angle  $\Theta_j = \arccos(\langle \hat{\mathbf{w}}_j, \mathbf{w} \rangle)$  between previous and possible directions, and the indicator function  $\mathbf{1}$ , which restricts the choice to angles below  $60^\circ$ . This assigns highest probabilities to directions which are most similar to the previous one. Up to an angle of  $30^\circ$ , directions are weighted almost equally, unless one of them matches better in terms of its volume fraction.

In line with the traditional UKF, our low-rank UKF simply takes a step along the chosen direction. We combine the joint low-rank method with a second-order Runge-Kutta integration scheme to increase precision in regions of high curvature. This involves selecting an initial direction, taking a tentative step with half the step size, repeating the steps from above with the interpolated fODF there, and using the resulting direction for the final step with the full step size.

In all cases, streamlines are terminated if the white matter density becomes too low or if we cannot find a valid direction within 60 degrees. Moreover, if the sum of angles over the last 30 mm of a streamline exceeds  $130^\circ$ , it is discarded as having unrealistically high curvature.

## 2.6. Data

We evaluate our two approaches with respect to each other and to the state of the art on three datasets. First, we investigate the benefit of spatial regularization when using all diffusion weighted volumes from the Human Connectome Project (Essen et al., 2013), or a small subset of them. Second, we compare our methods against those that participated in the 2015 ISMRM tractography challenge (Maier-Hein et al., 2017). Third, we provide qualitative results on clinical data of a pediatric tumor patient (Küpper et al., 2015).

### 2.6.1. HCP

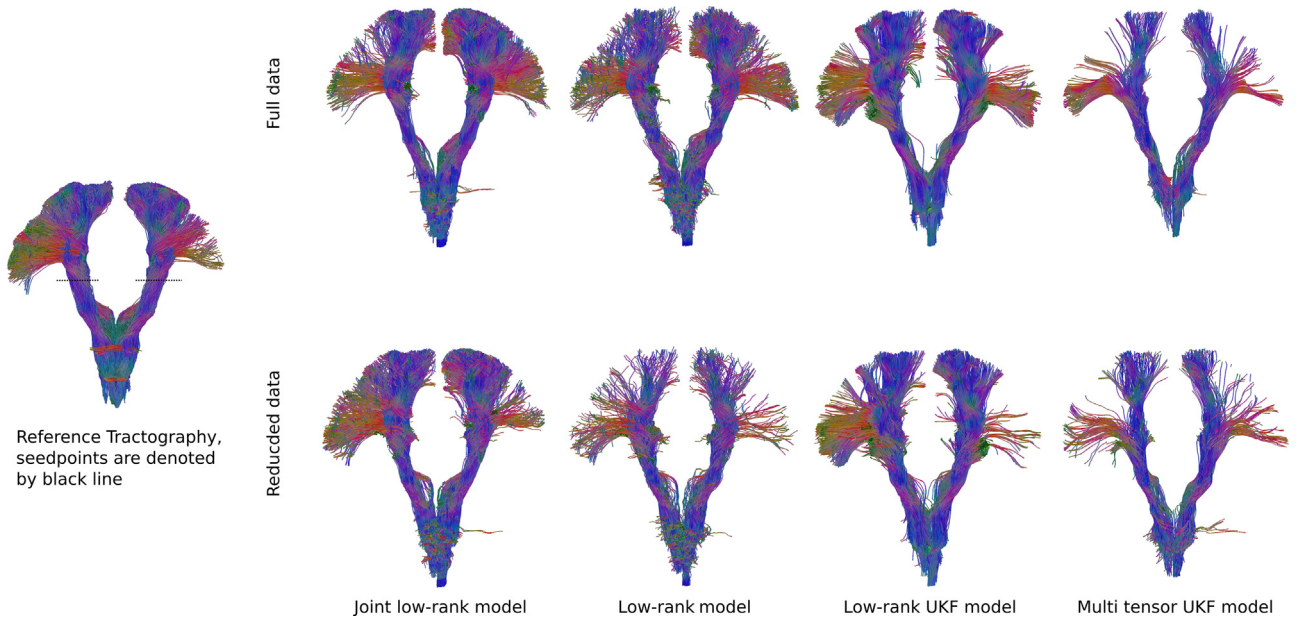
In the first experiment, we reconstruct 7 major white matter bundles within 11 subjects from the HCP dataset. That data has an isotropic resolution of 1.25 mm measured at b-values 1000, 2000, and 3000 s/mm<sup>2</sup> on 90 different directions each. Additionally, 18  $b = 0$  volumes were measured and the data was preprocessed as described by Essen et al. (2013).

We aim to reconstruct the tracts by following the protocols by Wakana et al. (2007), which involve ROI-based seeding, as well as inclusion and exclusion regions. As these protocols were designed for DTI tractography, we first adapted them to the needs of multi-fiber tractography based on a reference subject, and transferred them to all other subjects via linear image registration (Jenkinson et al., 2002). All streamlines which intersect any exclusion region or do not intersect all inclusion regions are removed.

Results are compared to reference reconstructions of major white matter bundles, which were created and published as part of the TractSeg project (Wasserthal et al., 2018). Seed points have been generated by intersecting the reference streamlines with a plane that intersects the streamlines at most once. An example is shown in the left of Fig. 2. We deal with seed points in crossing regions by initializing the previous tracking direction to the tangent of the reference streamline.

We investigated the extent to which spatial regularization improves tractography when using the full HCP data, and to which it helps to maintain feasibility of crossing-fiber tractography when reducing the





**Fig. 2.** Reconstruction of the corticospinal tract (CST) from the full HCP data (top), and from 30 gradient directions at  $b = 1000$  (bottom). The joint low-rank model permits the most complete reconstruction of the lateral spread, especially on the reduced data. It also leads to smoother streamlines than the unregularized low-rank model. Our low-rank variant of the UKF produces a more complete reconstruction than the traditional multi-tensor UKF.

amount of available data drastically, to a single  $b = 1000$  shell with only 30 gradient directions.

Results of our low-rank UKF are compared to the traditional multi-tensor UKF (Malcolm et al., 2010). To evaluate the benefit of our proposed modification as directly as possible, we use the above-described implementation in either case, and only replace the signal model. Parameters for the multi-tensor case are set as proposed by Malcolm et al., i.e., we set the measurement noise to  $\mathbf{Q}_{m_j} = 0.003$  and  $\mathbf{Q}_{\lambda_{ij}} = 25$ , and the process noise to  $\mathbf{R}_{ij} = 0.02$ .

Due to the probabilistic tracking, a density filter is used to remove isolated outliers from all results. For this, a density map is created by counting the number of streamlines which intersect with each voxel. This density map is binarized with a threshold and the streamlines are cut off at the first intersection with a low density region, starting from the seed point. We empirically set this threshold to 8 for the reference subject and map it to all other subjects by rescaling it with the seed ratio between the reference subject and the subject it is mapped to.

### 2.6.2. ISMRM

On the ISMRM 2015 challenge dataset, we compare our approaches to the 96 submissions made by 20 teams. The dataset is synthesized based on 25 major fiber bundles obtained from a HCP subject. It has an isotropic resolution of 2 mm, 32 gradient directions at a single b-value of  $1000 \text{ s/mm}^2$ , and two  $b = 0$  images with reversed phase-encoding.

We pre-processed the data with the MPPCA denoising algorithm (Veraart et al., 2016), Gibbs ringing removal (Kellner et al., 2016), and distortion correction with topup and eddy\_correct (Andersson et al., 2003). Then, we upsampled the data to 1 mm isotropic resolution and computed the fODFs. We obtained a white matter mask by thresholding the corresponding probability in a segmentation of the  $T_1$  weighted image (Zhang et al., 2001) at 0.4.

We seeded tractography twice at each voxel within the white matter mask, which resulted in around 0.8 million seeds, and kept streamlines with a minimum length of 40 mm. We used FFClust (Vázquez et al., 2020) to remove outliers by filtering out all streamlines which were not in clusters of at least three. We scored final results with a script that was provided by the challenge organizers after the challenge had been completed (<https://github.com/scilus/ismrm>).

### 2.6.3. Tumor

Finally, we demonstrate the benefit of tractography based on regularized low-rank tensor approximation on a clinical dataset of a child with a brain tumor (Küpper et al., 2015). This data has an isotropic resolution of 2.5 mm on  $96 \times 96 \times 60$  voxels measured at  $b = 3000 \text{ s/mm}^2$  along 60 gradient directions, as well as 3  $b = 0$  volumes. The data has been pre-processed with the standard FSL pipeline (Jenkinson et al., 2012). We reconstructed the CST by seeding in a plane and defining suitable inclusion and exclusion regions.

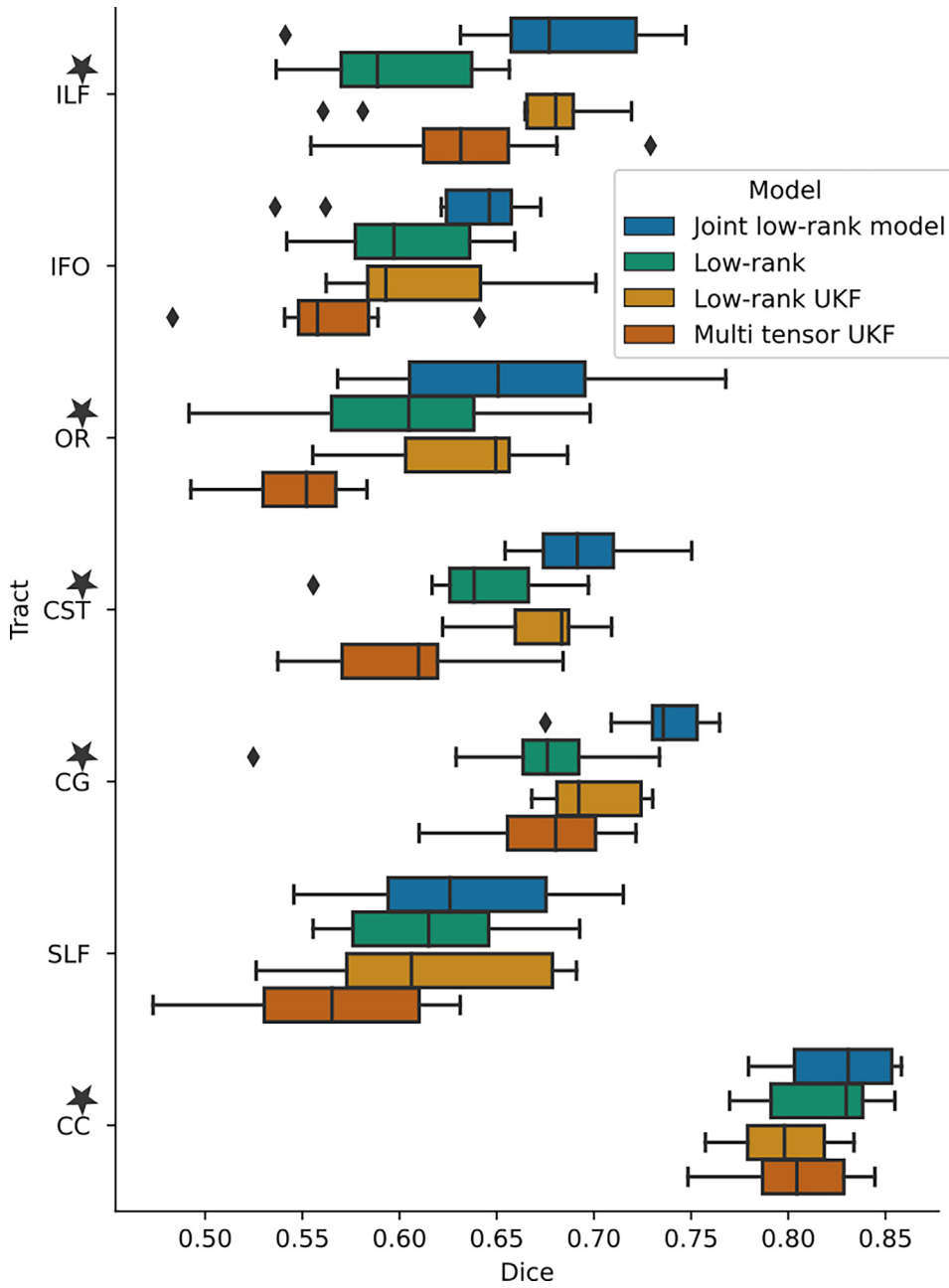
## 3. Results

### 3.1. Full and reduced HCP data

A qualitative comparison of reconstructing the corticospinal tract (CST) in an example subject is shown in Fig. 2. Compared to the traditional low-rank model, the joint approximation leads to more aligned streamlines on the full data (top row), and clearly increases the extent to which the CST can be recovered from the reduced data (bottom row). The low-rank UKF leads to even more organized streamlines, but misses parts of the lateral spread. The traditional multi-tensor UKF misses substantial parts even in the full data.

The visual impression that, in the reduced data, the joint low-rank approximation provides the most complete reconstruction, followed by the low-rank UKF, is confirmed by the quantitative results for all subjects in Fig. 3. It shows Dice scores achieved by different approaches for all seven tracts on the reduced dataset. The Dice score (Schilling et al., 2019) quantifies the spatial extent of the reconstructed tract, accounting both for over- and undersegmentation. It ranges from zero, if a tractography result does not intersect with the reference at all, to one, if the tractography covers the full reference, and nothing beyond. Results suggest that both our proposed innovations lead to an improvement compared to the respective baseline: The joint low-rank approximation achieves more accurate reconstruction than the traditional low-rank approach, and the low-rank UKF achieves better results than the multi-tensor UKF.

The outcome of a Friedman test ( $p \approx 1.9 \cdot 10^{-22}$ ) confirms a statistically significant difference between the models (Friedman, 1937). A paired, two sided Nemenyi post-hoc test (Nemenyi, 1963) indicates that all pairwise differences are significant ( $p < 0.05$ ), as shown in Table 1.



**Fig. 3.** Boxplots of the Dice scores of all subjects for all evaluated models, after reducing the HCP data to only 30 gradient directions at  $b = 1000$ . The joint low-rank model and low-rank UKF consistently achieve a greater median Dice than their respective baseline, with the exception of the CC, where low-rank and multi-tensor UKF produce similar numbers. The joint low-rank model attains greater median Dice than the low-rank UKF in all tracts, except for the ILF, where they are similar. Tracts in which paired rank differences of all four models are significant according to a Friedman test ( $p < 0.007$ , due to Bonferroni correction) are indicated by an asterisk.

**Table 1**

All  $p$  values from comparing different models in the reduced HCP dataset by a Nemenyi post-hoc test indicate significant differences in the ranking of Dice scores ( $p < 0.05$ ). Arrows denote which model led to greater Dice.

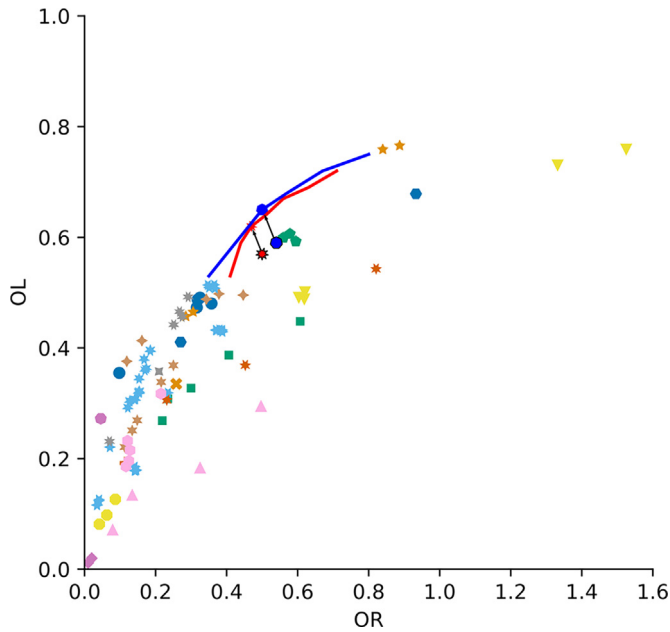
|                 | Low-rank | Multitensor UKF | Low-rank UKF |
|-----------------|----------|-----------------|--------------|
| Joint low-rank  | ← 0.001  | ← 0.001         | ← 0.001      |
| Low-rank        |          | ← 0.002         | ↑ 0.001      |
| Multitensor UKF |          |                 | ↑ 0.001      |

Inspecting the direction of the differences reveals that Dice scores from the joint low-rank approximation were higher than those from all other models. Additionally, the low-rank UKF yields higher Dice than both the low-rank approximation and the multi-tensor UKF. Finally, the traditional low-rank approximation exhibits higher Dice than the multi-tensor UKF.

Running times for generating 1000 streamlines on a single 3.3 GHz CPU core with our Python and Cython based implementation were 38.9 s for the traditional low-rank model, 43.5 s for the joint approximation, confirming that the overhead for regularization is moderate. Due to the high dimensionality of the measurement space, the multi-tensor UKF took 1212.0 s for the full HCP data. Our low-rank UKF reduced this to only 6.3 s. It is therefore not just significantly more accurate than the traditional UKF, but also represents the computationally most efficient alternative.

### 3.2. ISMRM challenge dataset

Figure 4 plots bundle overlap (OL) against overreach (OR) for all original participants of the ISMRM challenge and both of our approaches. Both are Pareto optimal: No challenge submission achieved lower OR with at least equal OL, nor higher OL without increasing OR. Moreover, both approaches simultaneously increase OL and reduce OR



**Fig. 4.** Evaluation on the 2015 ISMRM challenge in terms of overlap (OL) compared to overreach (OR). A high OL with a low OR is desirable. The low-rank approximation approach is marked with red stars, with an arrow pointing from the baseline (per-voxel optimization) to the proposed joint approximation. The UKF is marked with blue circles, with an arrow pointing from the baseline (multi-tensor) to the proposed low-rank UKF. Alternative operating points were obtained by varying white matter thresholds from 0.3 to 0.6 with a step size of 0.05, depicted by the blue and red curves. All other markers denote participants in the original challenge. Both proposed methods improve upon their respective baseline, and achieve higher OL than competitors with similar OR. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

compared to their respective baseline. In this experiment, the low-rank UKF achieved slightly higher OL than the joint approximation, at slightly higher OR. Our two proposed methods successfully reconstructed 23 out of 25 tracts. The low-rank UKF reconstructed 106 invalid bundles and achieved 43% valid connections, while the joint low-rank approach reconstructed 96 invalid bundles with 38% valid connections.

Considering that the white matter threshold plays a crucial role in regulating the OL to OR ratio we also present results for a range [0.3, 0.6] of reasonable thresholds, with step size 0.05. Results are shown as curves in Fig. 4. In this experiment, the low-rank UKF Pareto-dominated the joint low-rank model, by a small margin.

Figure 5 presents a qualitative comparison on the right cingulum, whose thin and curved parts present a challenge for tractography. Our joint low-rank approximation reconstructs the tract to a similar extent as traditional probabilistic tracking (challenge submission 2 from group 12 Maier-Hein et al., 2017), but its streamlines have a smoother visual appearance compared to the high-frequency oscillations in Fig. 5 (d). Traditional deterministic tracking (submission 0 from group 9) achieves similarly smooth curves, but misses part of the spatial extent of the bundle. The low-rank UKF produces the smoothest result, but is slightly less complete than the result in Fig. 5 (b).

### 3.3. Tumor data

Results are shown in Fig. 6, with the tumor depicted as a gray surface. The low-rank UKF in Subfigure (b) creates smooth and well-aligned streamlines, and is able to reproduce some of the lateral spread in the surrounding of the tumor. The joint low-rank approximation in Subfigure (a) creates even more spread, but also contains more outliers.

For comparison, Fig. 6 (c) shows the result from iFOD2, which is the default probabilistic tractography algorithm in MRtrix (Tournier et al., 2019) and produced the best results on this dataset in a previous study by Küpper et al. (2015). Both our proposed approaches are more sensitive with respect to reconstructing the lateral spread, both around the tumor and in the unaffected hemisphere.

To quantify this visual observation, we counted the number of lateral streamlines that reach the precentral cortex. We counted streamlines as being part of a lateral projection if the final ten integration steps (i.e., 7 mm) were more closely aligned with the left-right axis than with any other body axis. We employed FreeSurfer (Dale et al., 1999; Fischl et al., 2004) to generate a parcellation of the cerebral cortex and utilized the Möller-Trumbore algorithm (Möller and Trumbore, 1997) to identify lateral streamlines that intersect the part of the mesh representing the interface between the white matter and the precentral cortex. iFOD2 created only 16 corresponding streamlines, in contrast to 72 streamlines from the joint low-rank approximation and 122 streamlines from the low-rank UKF.

## 4. Discussion

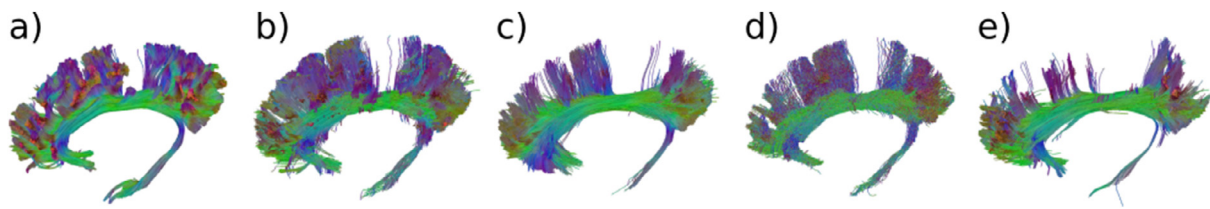
In this work, we have proposed two novel approaches to regularize diffusion MRI tractography. The first one introduces spatial neighborhood information into a previously proposed method that is based on low-rank approximations of higher-order tensor representations of fODFs. We have demonstrated that this has a positive impact on the reconstruction quality, at a minimal additional computational cost. The second approach replaced the traditional multi-tensor signal model in the UKF with a low-rank fODF approximation. This not only reduced the computational cost dramatically, but also increased the accuracy.

The UKF has originally been introduced for single-shell high angular resolution diffusion imaging (HARDI) data (Malcolm et al., 2010). We believe that its relatively weak result on the full HCP data (Fig. 2, top right) might indicate that its multi-tensor signal model is less suitable for multi-shell data. In particular, that model assumes that the dMRI signal from individual fiber compartments decays monoexponentially. Clear deviations from that assumption have been observed when measuring at more than a single high  $b$  value (Jensen et al., 2005). Estimating fODFs via a multi-shell multi-tissue extension of constrained spherical deconvolution (Jeurissen et al., 2014), or its generalization to arbitrary  $q$ -space samplings (Ankele et al., 2017), accounts for this. Another reason why it might lead to more accurate results is that it explicitly separates signal contributions from gray matter and cerebrospinal fluid.

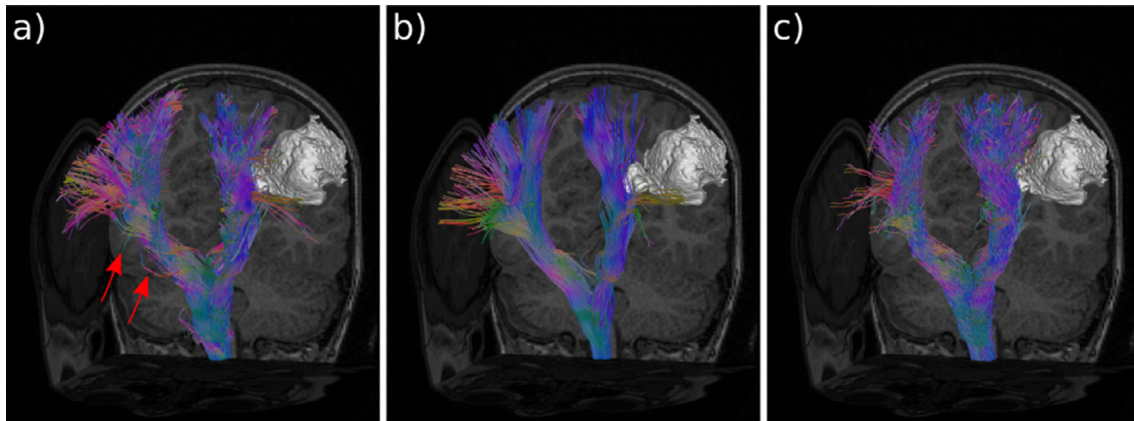
Experimental results on a strongly reduced subset of measurements in 11 HCP subjects indicate that spatial regularization permits a meaningful crossing fiber tractography even with only 30 gradient directions at  $b = 1000 \text{ s/mm}^2$ , with results that visually resemble those from the full data. Quantitatively, both methods significantly improve over their respective baselines, with the joint approximation leading to the highest Dice scores.

In this experiment, we compare our results to the same reference that was created for TractSeg (Wasserthal et al., 2018), a tractography method that uses neural networks to learn strong priors about the anatomy of specific bundles. Since our proposed methods do not use machine learning, we did not perform a direct comparison with TractSeg, and we do not expect to achieve higher accuracy. However, we do observe a strong continued use of traditional generic streamline-based tractography in the recent literature (Rousseau et al., 2022; Shastin et al., 2022). Therefore, we believe that improving such methods continues to be of interest and that, in addition to TractSeg's methodological merits, the data provided by that project are a useful resource in this context.

On the 2015 ISMRM tractography challenge, both proposed methods lead to higher overlap at reduced overreach compared to their baselines, and both are Pareto optimal with respect to all competitors from the challenge. Finally, a comparison on a clinical tumor dataset suggests



**Fig. 5.** Reconstruction of the right cingulum in the ISMRM challenge dataset using our proposed methods, and two submissions based on traditional probabilistic and deterministic approaches. a) Ground truth, b) Joint low-rank model, c) Low-rank UKF model, d) Probabilistic tracking, e) Deterministic tracking. Both proposed models achieve a more complete reconstruction than the deterministic method, while leading to smoother streamlines than the probabilistic one.



**Fig. 6.** Reconstruction of the corticospinal tracts from a clinical diffusion MRI scan of a tumor patient. The tumor segmentation is rendered as a gray surface. Both proposed methods (a) joint low-rank, b) low-rank UKF) permit a more complete reconstruction of lateral spread than MRtrix iFOD2, a widely used alternative c), and reconstruct more fibers around the tumor. The low-rank UKF b) generates smoother and better aligned streamlines than the joint low-rank method a), which creates more outliers, some of which are marked with red arrows. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

that our methods produce visually smooth streamlines that reconstruct larger parts of the lateral spread compared to a standard algorithm.

The small-scale fluctuations in traditional probabilistic tractography (Tournier et al., 2019), which are seen in Fig. 5 (d), reflect a random sampling of tracking directions from the fODF in each step. The smoother appearance of our streamlines results from computing a small number of potential tracking directions deterministically. Even though our methods are probabilistic as well, this is limited to selecting a direction in cases where multiple ones are similarly plausible.

When comparing the two proposed methods with each other, we note that the low-rank UKF is computationally more efficient and produces the visually smoothest results. On the other hand, the joint approximation led to a higher coverage of fiber spread when used with ROI-based seeding in the HCP experiment. We believe that increased computational effort and increased reliability in detecting secondary fiber compartments are both consequences of the iterative optimization of the joint low-rank approximation in each tracking step, while the UKF only updates its state once per tracking step. Despite this, the UKF led to slightly higher overlap with the reference tracts on the ISMRM challenge. We believe that the whole-brain seeding in that experiment compensated for the UKF's otherwise slightly lower sensitivity with respect to spread. Based on these considerations, we recommend the low-rank UKF for whole-brain tractography, the joint approximation for reconstructing the largest possible extent of a bundle from a single seeding region.

## 5. Conclusion

This work is the first to combine the ideas of low-rank higher-order tensor approximation and spatial regularization for diffusion MRI tractography. It introduced two novel algorithms for streamline-based tractography. The first one performs low-rank approximation jointly in a

local neighborhood, with a weighting scheme that ensures that regularization happens along tracts, and preserves tract boundaries. The second one integrates low-rank approximation into an unscented Kalman filter (UKF), which improved results and, at the same time, drastically reduced computational effort compared to traditional UKF tractography. Both proposed methods advance the state of the art in streamline-based dMRI tractography. In direct comparison, the joint approximation more fully recovered the spatial extent of spreading bundles from a single seed region, while the low-rank UKF achieved slightly higher overlap, at reduced computational cost, when used for whole-brain tractography.

## Data and Code Availability Statement

Code for the proposed tractography methods will be made available as part of the bonndit package (<https://github.com/MedVisBonn/bonndit>) at the time of publication. Data from the Human Connectome Project (<https://www.humanconnectome.org/>) and the 2015 ISMRM tractography challenge (<https://tractometer.org/ismrm2015/home/>) are freely available. The clinical data cannot be made openly available due to privacy issues.

## Ethics Statement

No data were newly acquired for the present study. Existing data have been collected in accordance with the ethical standards of the respective institutional and/or national research committees.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



## Credit authorship contribution statement

**Johannes Gruen:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Samuel Groeschel:** Resources, Validation. **Thomas Schultz:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition.

## Data availability

Links to code and data are provided in the data availability statement.

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